



CS 302: Paradigms of Programming

Course Code: CS 302

Course Name: Paradigms of Programming

L-T-P-C: 3-0-2-4

Prerequisites: IC 150 Computation for Engineers

Intended for: UG

Distribution: Compulsory for CSE; CS elective for EE and ME

Semester: 5th

Approval: 12th Senate (Wrong Credits: 0-0-3-2 were approved in 9th Senate)

L-T-P-C: 3-0-2-4

Approval: 12th Senate

Course Contents

- **Module I** Lambda Calculus - Syntax, Conversion, Reduction and Normal Order, Church-Rosser Theorem, Order of Evaluation, Currying, Integers, Booleans and Recursion.
- **Module II** Functional Programming - Scheme/Lisp syntax - expressions and functions/procedures, evaluation - naming, environment and the substitution model of function application, Higher-order functions and higher-order programming, Data abstraction.
- **Module III** Object-oriented Programming - Mutable data, modularity and state, Objects, Closures - data structures encapsulated into functions.
- **Module IV** Delayed Evaluation, Laziness and Infinite Data-structures - Stream Programming, Infinite streams, Streams as lazy Lists.
- **Module V** Logic Programming - Deductive Information retrieval, declarative programming - facts and rules, Search trees and Backtracking, Non-deterministic programming, Continuations and back-tracking, Prolog - arithmetic, recursion, cuts and negation, Real-life applications of Prolog. Prolog implementation in Scheme/Lisp - use of continuations and macros.